Figure Relay — one-page write-up draft

Status: preparation draft, not submitted. Results below are measured local runs; methods, per-run evidence, and limitations are in `docs/DELIVERY_VALIDATION.md`. Scripted teaching controls are labelled and excluded from these numbers.

See what survives when an AI reader reconstructs your scientific figure.

Scientific figures carry claims through arrows, geometry, conditions, axes, and visual grouping. An author can recognize the intended meaning even when a new reader cannot. Figure Relay makes an AI reader's interpretation inspectable through a reconstruction round trip.

The author provides a PNG/JPEG and confirms three to five atomic checkpoints. Reader A sees only the original image and produces a structured memory. A deterministic renderer receives only that persisted memory and produces a new raster figure. Reader B sees only the reconstruction. A separate comparison step examines the checkpoints against both memories, exposing first-read differences, reconstruction gaps, and subsequent reading differences. An optional direct-image checkpoint baseline is isolated from the main pipeline.

The application supports mechanism and structure diagrams and conceptual analysis plots through a shared representation of entities, relationships, conditions, uncertainty, axes, and facts. The report exposes the original and reconstruction, both memories, evidence, render audits, checkpoint-level findings, and a per-HTTP-attempt audit that separates nominal A/B/judge stages from retries and repairs. Exports and configuration fingerprints support review and controlled comparison after figure revision. Uncertain semantic matches require human review; no composite quality score is presented.

The prototype includes four attributed CC BY research figures spanning nonlinear physics, material lattice design, and two network-analysis plots, plus six paired engineering controls. Engineering validation comprises 75 passing unit tests across 11 files, clean lint and type checks, a successful production build, 24 history/deep-link browser checks plus 16 interface checks, and one real run driven end to end through the UI with no console errors.

Empirical validation uses a fixed protocol: three attempts per figure, every outcome recorded, failures never retried away, and one isolated direct-check baseline for every completed run (the four Reader A failures did not reach the baseline). With a local Qwen3-VL 4B instruct model (16k context, reader prompt 0.2.2), 8 of 12 runs completed: physics 2/3, materials 3/3, the metric/p-value plot 3/3, and a held-out network-permutation figure 0/3. All failures were an explicit Reader A stop with `incomplete` after exceeding the 10,000-token output limit; by label-level matching, no completed run lost a checkpoint marked confirmed in its run record. Those checkpoints were project-curated and marked confirmed by the protocol script; no independent item-by-item human confirmation record exists. Completed runs took 30–184 seconds. Two bounded prompt revisions were tested: 0.2.1 clarified "readable means complete" but caused a materials regression and was reverted; 0.2.2 clarified that `unsupported` means the image is unreadable, not that the paper's statistics fail. Neither change relaxed the completeness gate.

Retention by label is not scientific fidelity, and the reports now show it. On the saved metric/p-value run, the four earlier label-level checkpoints were retained, but Reader A sampled the flat `alpha = 0.05` reference line as y=0 and Reader B recorded it as (0,1)→(3,0). A project-curated, not author-confirmed check with a tolerance basis frozen from the source panel now judges both A and B contradicted; the earlier 0.05 tolerance that accepted A's y=0 is superseded. Deterministic checks take precedence over semantic model output, and the same audit flags that B moved "linear" into axis unit fields. The materials reconstruction reduced a lattice diagram to three labelled boxes, while confidence bands had no schema field and disappeared from the reconstruction. Small-magnitude comparisons scale their floating-point allowance to the operands rather than adding a fixed epsilon, so a 1e-10 A reference is not accepted as zero; B-only panels are listed as unmatched, not omitted. These are reasons the product reports which checks actually ran, attributes evidence to stages, and flags semantic matches for human review rather than scoring figure quality. All real runs recorded no explicit Reader A facts, so fact-by-fact comparison is explicitly marked "not executed"; an empty finding list is never presented as evidence of zero change.

The local application uses Next.js, a schema-validated model adapter, deterministic SVG-to-PNG reconstruction, and private filesystem run storage. Ollama with Qwen3-VL is the zero-API-fee local backend; hardware and download costs remain. No source caption or author checkpoint reaches either image reader.

The reconstruction is a diagnostic artifact, not a scientific truth oracle. The same model in independent requests can share errors, so A and B agreement is not statistical independence, and a reconstruction can introduce losses. The sample is small and does not establish cross-domain generalization or an advantage over direct checking.

Repository: https://github.com/snowgoose1975/Research-Figure-Evaluator (local implementation not yet pushed).